Hugo Marquerie 01/11/2025

Teo fn continua soporte compacto denso lp

Teorema 1. *$\mathcal{C}_c(\mathbb{R})$ es denso en $\mathcal{L}^p(\mathbb{R}, m)$ para todo $p \in [1, \infty)$.*

***Demostración*:** Sea $f \in \mathcal{L}^p(\mathbb{R}, m)$ y $\varepsilon > 0$. Por la `Prop-fn-simples-denso-lp/`??, tenemos que $\exists s \in \mathcal{S} \cap \mathcal{L}^p$ tal que $\|f - s\|_p < \varepsilon/2$. Si $s = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$, tomamos $M = \sum_{j=1}^n |a_j|$. Entonces, por el `Lem-aprox-indicatriz-continua-norma-lp/`Lema 1, $\forall j \in \mathbb{N}_n : \exists g_j \in \mathcal{C}_c(\mathbb{R}) : \left\|\mathbb{1}_{A_j} - g_j\right\|_p < \varepsilon/2M$. Sea $g = \sum_{j=1}^n a_j g_j \in \mathcal{C}_c(\mathbb{R})$, entonces

$$\|s - g\|_p = \left\|\sum_{j=1}^n a_j \left(\mathbb{1}_{A_j} - g_j\right)\right\|_p \leq \sum_{j=1}^n |a_j| \cdot \left\|\mathbb{1}_{A_j} - g_j\right\|_p \leq \sum_{j=1}^n |a_j| \cdot \frac{\varepsilon}{2M} = \frac{\varepsilon}{2}.$$

Ahora, por Minkowski, $\|f - g\|_p \leq \|f - s\|_p + \|s - g\|_p \leq \varepsilon/2 + \varepsilon/2 = \varepsilon$. ■

Referenciado en

- `Teo-fn-suave-soporte-compacto-denso-lp`

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